



CSC-361 – Machine Learning | Complex Computing Problem

Semester: Spring 2025

Track: Track B – Software-Based AI System

Max Marks: 4

Mapped CLO: CLO-1

Instructor: Dr. Shafiq Ur Rehman Khan

Track B – Software-Based AI System
Milestone 1 – Product Planning and System Design

Course: CSC-361 Machine Learning	Semester: Spring 2025
Track: Track B – Software-Based AI System	Due Date: 01 April 2025
Instructor: Dr. Shafiq Ur Rehman Khan	Max Marks: 4 / 20
Mapped CLO: CLO-1	

Mapped Course Learning Outcome (CLO)

- CLO-1: Explain the fundamental concepts of machine learning including supervised learning, unsupervised learning, and deep learning models.

Milestone Objective

Students will define the AI product, identify the target audience, design the system architecture, and plan the ML component. This milestone builds conceptual understanding of machine learning paradigms (supervised, unsupervised, deep learning) before implementation begins.

Required Deliverables

- Written Report (PDF, 3–4 pages)
- GitHub Repository link (if applicable)

Submission Requirements

- Problem definition with identified target audience
- System architecture diagram with labeled components
- Identification of ML paradigm (supervised/unsupervised/deep learning) with justification
- Dataset source and preliminary description
- Tools and technologies stack (Python, frameworks, libraries)
- Expected technical challenges

Evaluation Rubric — Total Marks: 4 / 20



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Group Name / No.:

Submission Date:

Criterion	Excellent (Full Marks)	Good (75%)	Satisfactory (50%)	Needs Work (25%)	Marks	CLO
Problem Definition & Target Audience	Well-defined problem with clearly identified user needs and scope	Problem defined; audience identified with limited detail	Problem vaguely defined; audience unclear	Problem not defined	0.75	CLO-1
System Architecture Diagram	Clear, labeled diagram showing data flow, ML component, and system layers	Diagram present but partially labeled	Rough diagram without clear flow	No diagram provided	0.75	CLO-1
ML Paradigm Identification	Correct paradigm (supervised/unsupervised/deep learning) identified with strong justification	Correct paradigm identified; justification limited	Paradigm identified but justification weak	No ML paradigm identified	1.0	CLO-1
Dataset Planning	Dataset fully identified: source, size, and relevance described; preprocessing needs noted	Dataset identified with partial description	Dataset mentioned briefly	No dataset information provided	0.75	CLO-1
Technical Planning (Tools & Challenges)	Complete technology stack listed; realistic challenges identified with mitigation ideas	Stack listed; challenges partially identified	Limited planning; generic challenges only	No planning provided	0.75	CLO-1

Marks Distribution Summary

Criterion	Max Marks	CLO
Problem Definition & Target Audience	0.75	CLO-1
System Architecture Diagram	0.75	CLO-1
ML Paradigm Identification	1.0	CLO-1
Dataset Planning	0.75	CLO-1
Technical Planning (Tools & Challenges)	0.75	CLO-1
Total	4.00	



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Important Notes: All group members must be able to explain their individual contribution during viva. Plagiarism in report or code will result in zero marks for the entire group. Accuracy alone is not an acceptable evaluation metric — use multiple metrics appropriate to the ML task.